



TALEEM CITY INSTITUTE

Ameenpur, Faisalabad

03126987979

Name:		Roll#:		Class:	Inter Part-I
Subject:	Physics-11	Date:		Time:	
Test Type #	Type 9 - Short Test (No Choice) - Marks=30				
Test Syllabus:	Unit-1,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i Which one of the following is the correct record for the diameter of wire when measured with a screw gauge of least count of 0.001 cm?
(A) 2.3 cm (B) 2.31 cm (C) 2.312 cm (D) 2.3124 cm
- ii The dimensions of gravitational constant G are:
(A) $[M^{-1}L^3T^{-2}]$ (B) $[ML^2T^{-2}]$ (C) $[M^2L^{-1}T^{-2}]$ (D) $[ML^{-2}T^{-1}]$
- iii Which pair has same dimensions?
(A) Work and power (B) Momentum and impulse (C) Force and torque (D) Torque and power
- iv The dimensions of volume flow rate of fluid are:
(A) $[LT^{-1}]$ (B) $[L^2T^{-2}]$ (C) $[L^3T^{-1}]$ (D) $[L^3T^{-2}]$
- v The dimension $[M^0 L T^0]$ represents the quantity:
(A) Length (B) Mass (C) Time (D) Velocity

Q.2 Write short answers of the following questions.

(8x2=16)

- i Define Precision. Which instrument can measure precise value meter rod or vernier caliper?
- ii How do you assess the total uncertainty in the final result for multiplication and division?
- iii If a precise measurement is also an accurate measurement. Explain your answer.
- iv The mass of a metal box measured by a lever balance is 2.2 kg. Two silver coins of masses 10.01 g and 10.02 g measured by a beam balance are added to it. What is now the total mass of the box correct upto the appropriate precision.
- v The diameter and length of a metal cylinder measured with the help of vernier callipers of least count 0.01 cm are 1.22 cm and 5.35 cm. Calculate the volume V of the cylinder and uncertainty in it.
- vi The wavelength λ of a wave depends on the speed v of the wave and its frequency ' f ' knowing that $[I] = [L]$, $[v] = [LT^{-1}]$ and $[f] = [T^{-1}]$. Decide which of these is correct? $f = v\lambda$ or $f = \frac{v}{\lambda}$.
- vii Write the dimension of pressure and density.
- viii Suppose, we are told that the acceleration of a particle moving in a circle of radius with uniform speed v is proportional to some power of r , say r^n , and some power of v , say v^m , determine the powers of r and v ?

NOTE: Attempt the long question.

(5+3=8)

- (b) Derive a relation for time period of a simple pendulum using dimension analysis. The various possible factors on which time period " T " may depend are: (i) mass of the body (ii) Length of the pendulum (iii) angle " θ " that thread makes with vertical (iv) acceleration due to gravity.

MCQs Ans Key.

Q:1 (C)

Q:2 (A)

Q:3 (B)

Q:4 (C)

Q:5 (A)



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Test Syllabus:	Unit-2,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i The magnitude of the resultant of two forces 6N and 8N acting at right angle is:
(A)6 N (B)10 N (C)14 N (D)16 N
- ii $\vec{A} \times \vec{B} = \vec{C}$ points along +z-axis, then the vector A and B must lie in
(A)yz-plane (B)zx-plane (C)xy-plane (D)xθ-plane
- iii As $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ hence scalar product is:
(A)associative (B)multiplicative (C)additive (D)commutative
- iv The magnitude of cross product of two parallel vectors $\vec{a}$ and $\vec{b}$ is equal to:
(A)ab cosθ (B)0 (C)2ab (D)1
- v Dimension of moment arm is:
(A)[M] (B)[T] (C)[LT] (D)[L]

Q.2 Write short answers of the following questions.

(8x2=16)

- i The vector sum of three vectors gives a zero resultant. What can be orientation of the vectors?
- ii How would the two vectors of same magnitude are oriented such that resultant vector has magnitude equal to each vector.
- iii What is the angle between two vectors of same magnitude? If they were to be combined to give a resultant equal to a vector of the same magnitude.
- iv A force of 10 N makes an angle of 60° with x-axis. Find its x and y components?
- v What do you know about Right Hand Rule? Also state it.
- vi If $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + \hat{k}$, then find $\vec{A} \cdot \vec{B}$.
- vii What is the moment of a force about the point lying on the axis of rotation?
- viii What is difference between moment arm and moment of force?

NOTE: Attempt the long question.

(5+3=8)

3(a) Explain the addition of vectors by rectangular components method.

- (b) A force $\vec{F} = 2\hat{i} + 3\hat{j}$ units, has its point of application moved from point A(1, 3) to the point B(5, 7). Find the work done.

MCQs Ans Key.

Q:1 (B)

Q:2 (C)

Q:3 (D)

Q:4 (B)

Q:5 (D)



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Test Syllabus:	Unit-3,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i A cricket ball is hit so that it travels straight up in air and it acquires 3 seconds to reach the maximum height. Its initial velocity is:
(A) 10 ms^{-1} (B) 12.2 ms^{-1} (C) 15 ms^{-1} (D) 29.4 ms^{-1}
- ii The linear momentum P of a body is given by:
(A) $P = m\omega$ (B) $P = r \times L$ (C) $P = mu$ (D) $P = m2u$
- iii To improve the jumping record, a long jumper should jump at an angle of:
(A) 30° (B) 45° (C) 60° (D) 90°
- iv Motion of a body along y-axis is:
(A) one dimensional (B) two dimensional (C) three dimensional (D) four dimensional
- v Height of projectile is maximum at an angle of:
(A) 45° (B) 60° (C) 30° (D) 90°

Q.2 Write short answers of the following questions.

(8x2=16)

- i An object is thrown vertically upward. Discuss the sign of acceleration due to gravity, relative to velocity, while the object is in air.
- ii Show that rate of change in momentum for an object is equal to applied force.
- iii Calculate the linear momentum of a ball of mass 100 gram which moves with 5 ms^{-1} along a straight line.
- iv Define isolated system with an example?
- v Differentiate between elastic and inelastic collision. Explain how would a bouncing ball behave in each case?
- vi Find the relation for force due to water flow.
- vii What is meant by projectile motion? Write range of the projectile?
- viii Define projectile motion. Derive expression for maximum height.

NOTE: Attempt the long question.

(5+3=8)

3(a) State and prove Law of conservation of Linear Momentum.

(b) Two spherical balls of 2.0 kg and 3.0 kg masses are moving towards each other with velocities of 6.0 ms^{-1} and 4 ms^{-1} respectively. What must be the velocity of the smaller ball after collision, if the velocity of the bigger ball is 3.0 ms^{-1} ?

MCQs Ans Key.

Q:1 (D)

Q:2 (C)

Q:3 (B)

Q:4 (A)

Q:5 (D)



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Test Type #	Type 9 - Short Test (No Choice) - Marks=30				
Test Syllabus:	Unit-4,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i If a body of mass 5kg is raised vertically through a distance of 1m, then work done is:
(A)49 J (B)4.9 J (C)490 J (D)0.49 J
- ii If the body of weight 'W' is lifted through a height 'h' then the work done will be
(A)-wg (B)Wh (C)Wh sinθ (D)zero
- iii If an agent consumes a power of 1kilo-watt for one hour, the work done is:
(A)one watt-hour (B)one watt (C)one kilo-watt hour (D)one mega watt hour
- iv The dimension of power is:
(A)MLT⁻¹ (B)ML²T⁻² (C)ML²T⁻¹ (D)ML²T⁻³
- v Kinetic energy of a body depends upon its:
(A)weight (B)density (C)velocity (D)acceleration

Q.2 Write short answers of the following questions.

(8x2=16)

- i In what situation work done by a force on a body is positive, negative or zero?
- ii A force 'F' acts through a distance 'L'. The force is then increased to '3F' and then acts through a further distance of '2L'. Draw the work diagram to scale and calculate total work done.
- iii Give name at least two conservative and two non-conservative forces.
- iv Name the four non-conservative forces.
- v Convert 1.4 kW into joule/sec.
- vi Write some methods to obtain solar energy.
- vii Define Absolute potential energy and write only its formula.
- viii A boy uses a catapult to throw a stone which accidentally smashes a green house window. List the possible energy changes.

NOTE: Attempt the long question.

(5+3=8)

3(a) What is a conservative force? Show that gravitational force is a conservative force.

(b) A man pushes a lawn mower with a 40 N force directed at an angle of 20° downward from the horizontal. Find the work done by the man as he cuts a strip of grass 20 m long.

MCQs Ans Key.

Q:1 (A)

Q:2 (B)

Q:3 (C)

Q:4 (D)

Q:5 (C)



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Test Type #	Type 9 - Short Test (No Choice) - Marks=30				
Test Syllabus:	Unit-5,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i A body rotating with angular velocity of 2 radian/s and linear velocity is also 2 ms^{-1} , then radius of circle is:
(A) 1 m (B) 0.5 m (C) 4 m (D) 2 m
- ii The rotational kinetic energy of a hoop of mass m moving down an inclined plane with velocity v will be:
(A) $\frac{1}{4}mv^2$ (B) $\frac{1}{2}mv^2$ (C) $\frac{3}{4}mv^2$ (D) mv^2
- iii Relation between the speed of disc and hoop at the bottom of an incline is:
(A) $v_{disc} = \sqrt{\frac{3}{4}}v_{hoop}$ (B) $v_{disc} = \sqrt{\frac{4}{3}}v_{hoop}$ (C) $v_{disc} = \sqrt{\frac{2}{5}}v_{hoop}$ (D) $v_{disc} = 2v_{hoop}$
- iv The force needed to kept the body into circular motion is called:
(A) Frictional force (B) Gravitational force (C) Centrifugal force (D) Centripetal force
- v Orbital velocity near surface of earth is given by:
(A) $\sqrt{2gR}$ (B) $\sqrt{\frac{2g}{R}}$ (C) $\sqrt{gR}$ (D) $\sqrt{\frac{g}{R}}$

Q.2 Write short answers of the following questions.

(8x2=16)

- i Prove that $a = r\alpha$
- ii Define centripetal force and centripetal acceleration.
- iii Explain what is meant by centripetal force and why it must be furnished to an object if the object is to follow a circular path?
- iv Why does the coasting rotating system slow down as some material object is added to the system during rotation?
- v A disc without slipping rolls down a hill of height 10.0 m. If the disc starts from rest at the top of hill, what is its speed at the bottom?
- vi What type of energies is possessed by a hoop moving down frictionless inclined plane?
- vii A disc is rolling down on an inclined plane. Find the relation for the speed of disc at its bottom.
- viii A body of mass " m " is suspended from the ceiling of an elevator. If the elevator is ascending with an acceleration " a ", what would be the value of " T " acting on the body?

NOTE: Attempt the long question.

(5+3=8)

3(a) What is meant by centripetal force? Derive relations for centripetal force and centripetal acceleration.

(b) The Moon orbits the Earth so that the same side always faces the Earth. Determine the ratio of its spin angular momentum (about its own axis) and its orbital angular momentum. (In this case, treat the Moon as a particle orbiting the Earth). Distance between the Earth and the Moon is $3.85 \times 10^8 \text{ m}$. Radius of the Moon is $1.74 \times 10^6 \text{ m}$.

MCQs Ans Key.

Q:1 (A)

Q:2 (B)

Q:3 (B)

Q:4 (D)

Q:5 (C)



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Test Syllabus:	Unit-6,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i The word FLUID means:
(A)To rise (B)To fall (C)To flow (D)To oppose
- ii For an ideal fluid in steady flow, the streamlines are:
(A)Parallel (B)Anti-parallel (C)Perpendicular (D)Intersect each other
- iii Laminar flow occurs at:
(A)High speed (B)Low speed (C)Zero speed (D)Very high speed
- iv A hose pipe ejects water at a speed of 0.3 ms^{-1} through a hole of area 10 cm^2 , flow rate will be:
(A) $3 \text{ m}^3 \text{ s}^{-1}$ (B) $3 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$ (C) $30 \text{ m}^3 \text{ s}^{-1}$ (D) $0.03 \text{ m}^3 \text{ s}^{-1}$
- v A 2 metre high tank is full of water. If a hole appears at its middle, then the speed of efflux is:
(A) 4.4 ms^{-1} (B) 10 ms^{-1} (C) 6.2 ms^{-1} (D) 20 ms^{-1}

Q.2 Write short answers of the following questions.

(8x2=16)

- i State Stoke's law and what are the limitation of this law?
- ii Why fog droplets appear to be suspended in air?
- iii State Bernoulli's relation for liquid in motion and describe its two applications.
- iv Explain the working of a carburetor of a motor car using Bernoulli's principle.
- v State Torricelli's theorem and write its relation.
- vi Give two applications of the Bernoulli's equation.
- vii Explain the term systolic and diastolic pressure.
- viii A person is standing nearby a fast moving train. Is there any danger that he will fall towards it?

NOTE: Attempt the long question.

(5+3=8)

3(a) Derive Torricelli's theorem and Venturi relation by applying Bernoulli's equation.

(b) What gauge pressure is required in the city mains for a stream from a fire hose connected to the mains to reach a vertical height of 15.0 m?

MCQs Ans Key.

Q:1 (C)

Q:2 (A)

Q:3 (B)

Q:4 (B)

Q:5 (A)



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Test Type #	Type 9 - Short Test (No Choice) - Marks=30				
Test Syllabus:	Unit-7,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i The acceleration of a projection on the diameter of a particle moving along a circle is:
(A) $\omega^2 x$ (B) ωx^2 (C) $-\omega^2 x$ (D) $-\omega x^2$
- ii The waveform of SHM is:
(A) Sine wave (B) Cosine wave (C) Tangent wave (D) Square wave
- iii If the time period of simple pendulum is 2 seconds, its frequency will be:
(A) 1 Hz (B) 0.5 Hz (C) 1.5 Hz (D) 2 Hz
- iv The time period of second pendulum is:
(A) 4 seconds (B) 3 seconds (C) 1 second (D) 2 seconds
- v P.E of a spring mass is stored in:
(A) Both mass and spring (B) Spring (C) Mass (D) None of the above

Q.2 Write short answers of the following questions.

(8x2=16)

- i Show that in simple harmonic motion, the acceleration is zero when the velocity is greatest and the velocity is zero when the acceleration is greatest.
- ii What happens to the period of the simple pendulum if the length is halved and mass of bob is doubled?
- iii What is meant by phase angle? Does it define angle between maximum displacement and the driving force?
- iv In an oscillating mass spring system if mass is doubled, how its time period will change?
- v Can we realize an ideal simple pendulum?
- vi What will be the potential energy of mass attached to a spring at amplitude of 5 cm, if its spring constant is 10 Nm^{-1} ?
- vii What is difference between free and forced oscillations?
- viii Describe the condition under which a vibrating body resonates with other body.

NOTE: Attempt the long question.

(5+3=8)

- 3(a) Derive the relations for time period, displacement and velocity of a mass spring system performing simple harmonic motion.
- (b) A spring, whose spring constant is 80.0 Nm^{-1} vertically supports a mass of 1.0 kg in the rest position. Find the distance by which the mass must be pulled down, so that on being released, it may pass the mean position with a velocity of 1.0 ms^{-1} .

MCQs Ans Key.

Q:1 (C)

Q:2 (A)

Q:3 (B)

Q:4 (D)

Q:5 (B)



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Test Syllabus:	Unit-8,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i The linear distance between two nearest points of a medium vibrating in phase is:
(A)Time period (B)Frequency (C)Phase difference (D)Wave length
- ii When sound waves enter in different medium, the quantity that remains unchanged is:
(A)Intensity (B)Speed (C)Frequency (D)Wave length
- iii Beats are produced due to:
(A)Super position of sound waves (B)Polarization of waves (C)Bernoullis effect
(D)Diffraction of sound waves
- iv Beats detectable easily upto frequency difference between two sound is:
(A)2 Hz (B)6 Hz (C)10 Hz (D)32 Hz
- v If a stationary wave is established along a stretched string of length " l " and it vibrates in one loop, the wave length is equal to:
(A) l (B) $\frac{l}{2}$ (C) $\frac{l}{3}$ (D) $2l$

Q.2 Write short answers of the following questions.

(8x2=16)

- i Explain why energy remains "standing" in the medium between nodes?
- ii Why sound travels faster in hydrogen than in oxygen?
- iii Write down the effects of variation of pressure and density on the speed of sound in a gas.
- iv What are the factors upon which speed of sound in air depends?
- v What happened when a jet plane like a Concorde flies faster than speed of sound?
- vi Why sound travels faster in hydrogen than in oxygen?
- vii As a result of a distant explosion, an observer senses a ground tremor and then hears the explosion. Explain the time difference.
- viii What changes are observed if a wave is reflected from a denser medium?

NOTE: Attempt the long question.

(5+3=8)

3(a) Define stationary waves and show that the stationary waves in a stretched string are harmonic series.

(b) Two tuning forks exhibit beats at a beat frequency of 3 Hz. The frequency of one fork is 256 Hz. Its frequency is then lowered slightly by adding a bit of wax to one of its prong. The two forks then exhibit a beat frequency of 1 Hz. Determine the frequency of the second tuning fork.

MCQs Ans Key.

Q:1 (D)

Q:2 (C)

Q:3 (A)

Q:4 (C)

Q:5 (D)



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Test Syllabus:	Unit-9,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i The locus of all points in a medium having the same phase of vibration is called:
(A)Crest (B)Trough (C)Wavelength (D)Wave front
- ii Michelson's interferometer's can be used to find the:
(A)Velocity of light (B)Velocity of sound (C)Wavelength of light (D)Wavelength of sound
- iii If blue light is used as compare to red light then fringe spacing:
(A)Increases (B)Decreases (C)Remain same (D)Becomes zero
- iv The equation for michelson's interference is:
(A) $L = 2m\lambda$ (B) $L = m\lambda/2$ (C) $\lambda L = 2m$ (D) $\lambda L = m/2$
- v Michelson's interferometer can be used to find the:
(A)Velocity of light (B)Wavelength of light (C)Velocity of sound (D)Wavelength of sound

Q.2 Write short answers of the following questions.

(8x2=16)

- i What is meant by wavefront? Give its types.
- ii How does one can obtain a plane wave?
- iii On what factors, the distance between adjacent bright fringes in Young's Double Slit Experiment depends?
- iv How will you increase the fringe width in young's double slit experiment?
- v How the distance between interference fringes will be affected if the distance between the slits in the young's experiment is doubled?
- vi In Newton's ring, why are the fringes circular?
- vii Describe the construction of Michelson's interferometer with the help of diagram.
- viii What is meant by the diffraction of light?

NOTE: Attempt the long question.

(5+3=8)

3(a) Explain Young's Double slit experiment to study the phenomenon of interference of light.

(b) An X-ray beam of wavelength λ undergoes a first order reflection from a crystal when its angle of incidence to a crystal face is 26.5° , and an X-ray beam of wavelength 0.097 nm undergoes a third order reflection when its angle of incidence to that face is 60.0° . Assuming that the two beams reflect from the same family of planes, calculate (a) the interplanar spacing of the planes and (b) the wavelength λ .

MCQs Ans Key.

Q:1 (D)

Q:2 (C)

Q:3 (B)

Q:4 (B)

Q:5 (B)



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Test Syllabus:	Unit-10,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i If the object is at 5 cm from the lens of simple microscope than its magnifying power will be:
(A)5 (B)10 (C)15 (D)25
- ii In compound microscope, the focal length of objective is:
(A)Small (B)Large (C)Very large (D)Both (A) and (B)
- iii Which of the following gives erect image:
(A)Simple microscope (B)Galilean telescope (C)Terrestrial telescope (D)All the above
- iv If focal length of objective and eye piece is 0.5 m and 10 cm respectively then magnifying power of telescope will be:
(A)5 (B)0.5 (C)10 (D)20
- v If $f_o = 100$ cm; $f_e = 5$ cm length and magnifying power of an astronomical telescope is:
(A)0.05 cm ; 20 (B)95 cm ; 20 (C)20 cm ; 500 (D)105 cm ; 20

Q.2 Write short answers of the following questions.

(8x2=16)

- i Find magnifying power of convex lens 25 cm focal length acts as a magnifying glass.
- ii Describe with the help of ray diagrams, how a single-biconvex lens can be used as a magnifying glass?
- iii How can the resolving power of compound microscope be increased?
- iv What do you mean by "Normal adjustment" in an astronomical telescope?
- v If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens?
- vi Define critical angle and total internal reflection.
- vii Define the critical angle.
- viii How the light signal is transmitted through the optical fibre?

NOTE: Attempt the long question.

(5+3=8)

- 3(a) What is compound microscope draw rays diagram and derive expression for the magnification of compound microscope?
- (b) A telescope objective has focal length 96 cm and diameter 12 cm. Calculate the focal length and minimum diameter of a simple eye piece lens for use with the telescope, if the linear magnification required is 24 times and all the light transmitted by the objective from a distant point on the telescope axis is to fall on the eye piece.

MCQs Ans Key.

Q:1 (A)

Q:2 (A)

Q:3 (D)

Q:4 (A)

Q:5 (D)



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Test Syllabus:	Unit-11,				

Q.1 Circle the Correct Answers.

(6x1=6)

- i A device based upon the thermodynamics property of matter is called:
(A)Calorimeter (B)Heat engine (C)Thermometer (D)Voltmeter
- ii A diatomic gas molecules has:
(A)Translational energy only (B)Rotational energy only
(C)Vibrational energy only (D)All Translational, Rotational and Vibrational energy
- iii A system does 600 J of work and at the same time has its internal energy increased by 320 J. How much heat has been supplied?
(A)280 J (B)920 J (C)600 J (D)200 J
- iv The temperature of a system remains constant in:
(A)Adiabatic process (B)Isobaric process (C)Isothermal process (D)Adiabatic process
- v A heat engine operates between the temperature 1000 K and 400 K. Its efficiency is:
(A)100% (B)70% (C)60% (D)50%

Q.2 Write short answers of the following questions.

(8x2=16)

- i Define pressure of gas.
- ii Why should chimney be tall for its better working?
- iii Can the mechanical energy be converted completely into heat energy? If so give an example.
- iv What is adiabatic process? Under what conditions this process occurs?
- v Write two limitations of first law of thermodynamics.
- vi Explain bicycle pump as an example of first law of thermodynamics.
- vii Is it possible to construct a heat engine that will not expel heat into the atmosphere?
- viii State Carnot Theorem:

NOTE: Attempt the long question.

(5+3=8)

- 3(a) What is a Carnot heat engine? Show that efficiency of a Carnot heat engine depends on the temperature of the hot and cold reservoirs.
- (b) Show that ratio of the root mean square speeds of molecules of two different gases at a certain temperature is equal to the square root of the inverse ratio of their masses.

MCQs Ans Key.

Q:1 (B)

Q:2 (D)

Q:3 (B)

Q:4 (C)

Q:5 (C)